

Kasra Sinaei

Robotic Software Developer | Control & Physical AI Researcher | Authorized to work in the US
[kassra-sinaei.github.io](https://github.com/kassra-sinaei) | kasra.sinaei@gmail.com | (582)203-9717 | kasra@psu.edu

EDUCATION

PENNSYLVANIA STATE UNIVERSITY

PHD IN ELECTRICAL ENGINEERING
 Thesis: Safety Guarantee for Learning Policies via Barrier Functions
 May 2027 | State College, PA

PENNSYLVANIA STATE UNIVERSITY

MSC. IN ELECTRICAL ENGINEERING
 Paper: Model-free Safety-critical Position Control of Robots
 May 2024 | State College, PA

UNIVERSITY OF TEHRAN

BSC. IN MECHANICAL ENGINEERING
 Capstone Project: Motion Planning & Control of Wheeled-legged Robot
 Jan 2022 | Tehran, Iran
 Cum. GPA: 17 / 20

SKILLS

PROGRAMMING

C++ • C • Python • MATLAB • Java • C# • Shell

SOFTWARE DEVELOPMENT

ROS1/2 • Git • Docker • CUDA • .NET • AWS •
 Kubernetes • Web Stack • Android Studio

ROBOTICS & SIMULATION

Nvidia IsaacSim • MuJoCo • Gazebo • PyBullet •
 Pinocchio • Moveit • ROS Nav

COMPUTER VISION & AI

PyTorch • OpenCV • CUDA • Hugging-face •
 Transformers • Lerobot • YOLO • W&B

THEORETICAL

Nonlinear Control • Convex Optimization • Adaptive/Robust Control • VLM/ VLA • RL • SLAM •
 Motion Planning • MPC • Trajectory Optimization

CAD/CAE

SolidWorks • ANSYS • CATIA • AutoCAD

TEACHING ASSISTANT

Intro. to Linear Control Systems (EE380)	SP-2026
Continuous-Time Linear Systems (EE350)	FA-2025
Intro. to Digital Control Systems (EE482)	SP-2024
AI and ML in Mechanical Engineering	FA-2021
Electrical Circuits and Electrical Machines	FA-2020

LANGUAGE SKILLS

ENGLISH

Full Professional Proficiency C2 (CEFR)

SPANISH

Elementary Proficiency A2 (CEFR)

EXPERIENCE

CHEWY CO. | AI RESEARCH SCIENTIST INTERN

Jun 2026 - Sep 2026 | Boston, MA
 Contributing to Chewy's **fleet orchestration** layer (RMS), which couples decision-making at scale, shared spatial understanding, and safety/resilience for **multi-robot** operation in dynamic, human-occupied environments.

ALPHAZ INC. | ROBOTICS SOFTWARE DEVELOPER INTERN

May 2025 - Aug 2025 | Pasadena, CA
 Deployed **trajectory planning** algorithm for dual-arm **mobile manipulator** in real-time C++. Built high-reliability ROS2 **perception pipelines** (C++/Python) for state estimation and object detection. Integrated research code into production navigation stack.

RESEARCH

CARL | GRADUATE RESEARCHER

Aug 2022 - Now | State College, PA
 developing **theoretical guarantees** for safety-critical control systems and implementing model-free **safety** filters by bridging the gap between classical **control theories** and modern **foundation model policies**. Validated the proposed methods in simulation and hardware tests.

CAST | UNDERGRADUATE RESEARCHER

Sep 2019 - Mar 2022 | Tehran, Iran
 Developed core software for full-size **humanoid robot** at CAST. Implemented **low-level drivers**, **state estimation** fusing IMU/encoders, and closed-loop balance controllers. Designed gait controllers for traversing uneven terrain.

AWARDS & HONORS

MELVIN P. BLOOM MEMORIAL GRADUATE FELLOWSHIP | \$850 | 2024

Awarded to outstanding graduate students in Electrical Engineering based on academic excellence and research potential.

MILTON & ALBERTHA LANGDON MEMORIAL GRADUATE FELLOWSHIP | \$810 | 2023

Merit-based fellowship recognizing exceptional graduate student performance and research contributions.

DR. ARTHUR H. WAYNICK GRADUATE SCHOLARSHIP IN EE | \$3,147 | 2022

Prestigious scholarship awarded to top-performing graduate students in Electrical Engineering department.

COLLEGE OF ENGINEERING GRADUATE FELLOWSHIP | \$3,000 | 2022-2023

Competitive fellowship supporting outstanding graduate students pursuing advanced degrees in engineering disciplines.

GOVERNMENTAL BSC. SCHOLARSHIP | FULL TUITION WAIVER | 2017-2021

National merit-based scholarship covering complete undergraduate tuition for academically distinguished students.

PUBLICATIONS

- [1] **K. Sinaei**, H.-C. Wu, and D. Ebeigbe, **Safe vision language action model via barrier enhanced flow matching**, *IEEE Robotics and Automation Letters*, 2026.
- [2] **K. Sinaei** and D. Ebeigbe, **Safe adaptive control with vanishing conservativeness for robotic systems with unknown dynamics via barrier functions**, *IEEE Robotics and Automation Letters*, 2026.
- [3] **K. Sinaei**, K. Weerakoon, C. Bradley, S. A. Fakoorian, and D. Ebeigbe, **Motion planning for mobile manipulators navigating doorways via model predictive path integral**, *ASME Letters in Dynamic Systems and Control*, 2026.
- [4] **K. Sinaei** and D. Ebeigbe, **Safe robust control of nonlinear systems with uncertain regressor and parameter vector**, in *2025 IEEE Conference on Control Technology and Applications (CCTA)*, 2025.
- [5] **K. Sinaei**, H.-C. Wu, and D. Ebeigbe, **Safety-critical position control of robots: A model-free approach**, in *2025 American Control Conference (ACC)*, 2025.
- [6] P. Abdolhazehzad, A. Yousefi-Koma, A. Vedadi, **K. Sinaei**, B. Maleki, and M. Shafiee, **Online bipedal locomotion adaptation for stepping on obstacles using a novel foot sensor**, in *2022 IEEE-RAS 21st International Conference on Humanoid Robots (Humanoids)*, IEEE, 2022.
- [7] A. Vedadi, **K. Sinaei**, P. Abdolhazehzad, S. S. Aboumasoudi, and A. Yousefi-Koma, **Bipedal locomotion optimization by exploitation of the full dynamics in dcm trajectory planning**, in *2021 9th RSI International Conference on Robotics and Mechatronics (ICRoM)*, IEEE, 2021.
- [8] **K. Sinaei** and M. Yazdi, **Pid controller tuning with deep reinforcement learning policy gradient methods**, in *Proceedings of the 29th International Conference of Iranian Society of Mechanical Engineers*, Tehran, Iran, 2021.

SELECTED PROJECTS

More details on open source projects available at kassra-sinaei.github.io.

SAFE VLA VIA BARRIER ENHANCED FLOW MATCHING | THESIS RESEARCH

2026 | CARL, Penn State University

Developing **vision-language-action (VLA)** models for robotic manipulation with safety guarantees. Modified the base Flow Matching algorithm to effectively bridge **foundation models** and CBF to ensure safe deployment of learned policies. Validated through hardware experiments on manipulation tasks.

AUTONOMOUS DOOR OPENING WITH MOBILE MANIPULATORS | INTERNSHIP PROJECT

2025 | AlphaZ Inc.

Developed a fully autonomous system for door handle **detection** and **manipulation** using mobile manipulator. Implemented a novel motion planning algorithm, **real-time** control and **perception** pipelines.

MODEL-FREE SAFETY-CRITICAL ROBOT CONTROL | MASTER'S THESIS PROJECT

2022 - 2024 | Carl, Penn State University

Created a model-free safety control framework using **Control Barrier Functions** for position controlled robotic systems. Validated safety guarantees through **Lyapunov** analysis and **Set invariance** principles.

AUTONOMOUS RACING: OPTIMAL RACING LINE & VEHICLE CONTROL | COURSE PROJECT

2023-2024 | Penn State University

Utilized **RL** policy (PPO, SAC) for autonomous racing in simulation. Created **training pipeline** with reward shaping, episode logging, and performance metrics. Compared **RL** agents with **MPC** baselines.

HUMANOID ROBOT GAIT CONTROLLER | UNDERGRADUATE CAPSTONE PROJECT

Jan 2020 - May 2022 | University of Tehran

Developed **state estimator** and closed-loop **balance controllers** for full-size **humanoid**. Implemented adaptive gait patterns for traversing uneven terrain under uncertainty.

DEEP REINFORCEMENT LEARNING FOR PID TUNING | INDEPENDENT RESEARCH

2021 | University of Tehran

Implemented **RL** algorithms (PPO, A3C) for automatic PID controller gain optimization. Built **training framework** with custom gym environments and reward engineering pipelines.

CETIFIED TRAININGS

- NVIDIA DLI: Fundamentals of Deep Learning
- AWS Certified Cloud Practitioner
- ROS Industrial Training
- Advanced Programming with CPP